Sining Gong

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Computational Mathematics, Science and Engineering
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Professional Experience

Aug. 2023–present	**Postdoctoral Associate**, Michigan State University Center of Hierarchical and Robust Modeling for Non-Equilibrium Transport (CHaRMNET)
Sept. 2018– May 2023	**Graduate Student Researcher**, Brown University Division of Applied Mathematics
May 2016– May 2017	**Team Leader**, Beijing Normal University National Undergraduate Training Program for Innovation and Entrepreneurship Project: *Tracking Targets by Particle Filters Based on Color Histogram*

Education

2018–2023	**Ph.D. in Applied Mathematics**, Brown University, Providence, RI Dissertation: *Applications of FEM with Macro-Elements: Discrete Elasticity Sequences and Convergence of Lagrange Elements for a Maxwell Eigenvalue Problem* Advisor: Johnny Guzmán
2014–2018	**B.S. in Mathematics and Applied Mathematics**, Beijing Normal University, Beijing, China **Minor: Psychology** Thesis: *The Green's Function for the Helmholtz Equation on Special Domains* Advisor: Haigang Li

Research Interests

- High-order, structure-preserving numerical methods for fluid and kinetic equations
- Low-rank tensor approximations and structure-preserving machine learning
- Finite element exterior calculus and spline theory
- Method of lines transpose (kernel-based operators) and particle-in-cell methods
- WENO/ENO methods, finite element methods, and computational fluid dynamics

Honors and Awards

2023	**Stella Dafermos Award**, Brown University Awarded to outstanding graduate students in honor and memory of Stella Dafermos.
June 2020	**Graduate School Summer Fellowship**, Brown University
June 2018	**Outstanding Graduate Award**, Beijing Normal University
June 2018	**Departmental Outstanding Thesis Award**, Beijing Normal University
2015, 2016, 2017	**First Prize Scholarship**, Beijing Normal University

Journal Articles (Appeared/Accepted/Submitted)

1. Christlieb, A., **Gong, S.**, Qiu, J., & Zheng, N. A Sampling-Based Adaptive-Rank Approach to the Wigner–Poisson System. *SIAM Journal on Scientific Computing*, 2026. DOI: 10.1137/25M1776019.
2. Padilla-Gomez, F. A., **Gong, S.**, Murillo, M. S., Graziani, F. R., & Christlieb, A. Quantum Kinetic Modeling of KEEN waves in a Warm-Dense Regime. *Physics of Plasmas*, 2026. DOI: 10.1063/5.0308425.

3. Christlieb, A., **Gong, S.**, & Yang, H. Boundary corrections for kernel approximation to differential operators. *Journal of Scientific Computing*, 2025. DOI: 10.1007/s10915-025-03083-w.
4. **Gong, S.**, Guzmán, J., & Neilan, M. A Note on the Shape Regularity of Worsey–Farin Splits. *Journal of Scientific Computing*, 2023. DOI: 10.1007/s10915-023-02159-9.
5. Boffi, D., **Gong, S.**, Guzmán, J., & Neilan, M. Convergence of Lagrange finite element methods for Maxwell eigenvalue problem in 3D. *IMA Journal of Numerical Analysis*, 2023. DOI: 10.1093/imanum/drad053.
6. **Gong, S.**, Gopalakrishnan, J., Guzmán, J., & Neilan, M. Discrete elasticity exact sequences on Worsey-Farin splits. *ESAIM: M2AN*, 2023. DOI: 10.1051/m2an/2023084.
7. Christlieb, A., Chacón, L., & **Gong, S.** An Energy-Conserving Unstaggered Electromagnetic-Potential Particle-in-Cell Method, Part I: Non-relativistic Generalized-Momentum Formulation. **Submitted** to *SIAM Journal on Scientific Computing*. arXiv:2606.15035.
8. Christlieb, A., **Gong, S.**, Qiu, J., & Zheng, N. A Structure-preserving Adaptive-Rank Approach to the High-Dimensional Wigner-Poisson System. **Submitted** to *SIAM Journal on Scientific Computing*. arXiv:2606.15067.
9. Christlieb, A., **Gong, S.**, Padilla-Gomez, F. A., & Qiu, J. A Conservative Adaptive Rank Method for the Wigner-Poisson System. **Submitted** to *Journal of Scientific Computing*. arXiv:2606.20234.
10. Christlieb, A., Chacón, L., & **Gong, S.** An Energy-Conserving Unstaggered Electromagnetic-Potential Particle-in-Cell Method, Part II: Fully Relativistic Formulation with an Adapted Higuera–Cary Pusher. **Submitted** to *SIAM Journal on Scientific Computing*. arXiv:2609.36383.
11. Christlieb, A., **Gong, S.**, Huang, J., Krupansky, N. A., & Zhou, S. Hyperbolic Machine Learning Moment Closures for the Vlasov–Poisson–BGK Equations. **Submitted** to *Journal of Scientific Computing*.

Manuscripts in Preparation

1. Christlieb, A., **Gong, S.**, Graziani, F. R., Murillo, M. S., & Padilla-Gomez, F. A. Quantum-Correct Bhatnagar-Gross-Krook operator for coupling collisions to the Wigner-Poisson model.
2. Christlieb, A., **Gong, S.**, & White, S. R. A Real-Space Kernel-Based Vector Calculus: Part I – An $O(N)$ Unconditionally Stable Maxwell Solver without Spatial Staggering.
3. Christlieb, A., **Gong, S.**, Gottlieb, S., & Grant, Z. J. A Fully-implicit High order Strong Stability Preserving Method with Asymptotic-Preserving Property.

Selected Presentations

Selected from more than 25 presentations.

Oct. 2026	*A Structure-preserving Adaptive-Rank Approach to the High-Dimensional Wigner-Poisson System.* Invited talk, Numerical Analysis & PDE Seminar, University of Delaware.
July 2026	*A Structure-preserving Adaptive-Rank Approach to the High-Dimensional Wigner-Poisson System.* SIAM Annual Meeting.
Nov. 2025	*A Sampling-Based Adaptive Rank Approach to the Wigner-Poisson System.* SIAM NNP 2025.
Aug. 2025	*A Kernel-Based Time-Stepping Framework for Stiff PDEs: Revisiting MOLT with SSP-RK.* Innovative and Efficient Strategies for Stiff Differential Equations workshop, ICERM.
July 2025	*A Sampling-Based Adaptive Rank Approach to the Wigner-Poisson System.* 15th International Conference on Spectral and High Order Methods (ICOSAHOM 2025).

July 2024	*High order unconditionally strong stability preserving (SSP) implicit two-derivative Runge–Kutta schemes and their extensions.* SIAM Annual Meeting.
Feb. 2023	*Convergence of Lagrange FEM for Maxwell Eigenvalue Problem in 3D.* Pitt AWM Student Seminar, University of Pittsburgh.

Professional Service

Journal Reviewing

- *Journal of Computational Physics* (2026)
- *Computer Physics Communications* (2026)
- *Computer-Aided Design* (2025)
- *Communications on Applied Mathematics and Computation* (2025)
- *Computers & Mathematics with Applications* (2024; two manuscripts)
- *Results in Applied Mathematics* (2024)

Conference Organization

Organizer, "Advances in Particle Methods for High-Dimensional Kinetic Equations," minisymposium, SIAM CSE27 (approved; forthcoming, 2027).

Teaching Experience

Michigan State University

Fall 2026	**Instructor**, MTH 850: Numerical Analysis I (graduate course)
Fall 2025; Spring 2024	**Instructor**, MTH 132: Calculus I (undergraduate course; 90 students)

Brown University

Spring 2023	**Teaching Assistant**, APMA 1160: An Introduction to Numerical Optimization
Fall 2022	**Teaching Assistant**, APMA 1690: Computational Probability and Statistics
Summers 2022 and 2019	**Teaching Assistant**, Summer Review Course for Analysis
Spring 2020	**Head Teaching Assistant**, APMA 340: Methods of Applied Mathematics II
Fall 2019	**Teaching Assistant**, APMA 330: Methods of Applied Mathematics I

Mentoring Experience

2024–present	**Francisco Alejandro Padilla Gomez**, Ph.D. student of Prof. Andrew Christlieb, Michigan State University Research mentoring on Wigner-Poisson solvers for KEEN waves and the development of conservative Wigner-Poisson solvers.
May 2025–May 2026	**Doyoung Wang**, undergraduate student, Michigan State University Provided guidance on the Schrödinger and Wigner-Poisson equations and the implementation of Wigner-Poisson solvers.

Programming, Research Software, and Languages

Programming and scientific computing: MATLAB, FEniCS, Python, and LaTeX (extensive experience); C++ (research implementations); C, Mathematica, and Julia (some experience).
Selected research code: github.com/sgong11

- **WignerPoissonAdaptiveRank**: MATLAB implementations of full-rank and adaptive-rank Wigner–Poisson solvers supporting the published paper.
- **MOLt_boundary**: Tutorials in Jupyter notebooks for the method of lines transpose framework.
- **Unstaggered_PIC**: C++17 reference implementations of energy-conserving, generalized-momentum particle-in-cell methods.

Languages: Mandarin (native); English (fluent).